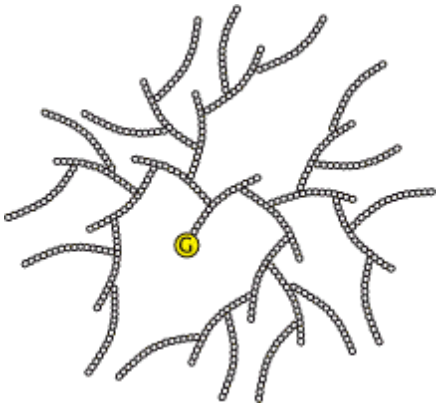
Glycogen Metabolism



PART-2: GLYCOGEN DEGRADATION

LEARNING OBJECTIVES

At the end of the lecture, students should be able to:

- Enlist enzymes required for glycogenolysis
- Describe steps of glycogen degradation
- Discuss the role of PLP in glycogenolysis
- Describe in detail how glycogenolysis is regulated by allosteric and covalent modification.
- Discuss the role calcium in regulation of glycogen breakdown.

Glycogenolysis (Degradation of glycogen)

Glycogen degradation consists of three steps:

(1) The release of glucose 1-phosphate from glycogen,

(2) The remodeling of the glycogen substrate to permit further degradation, and

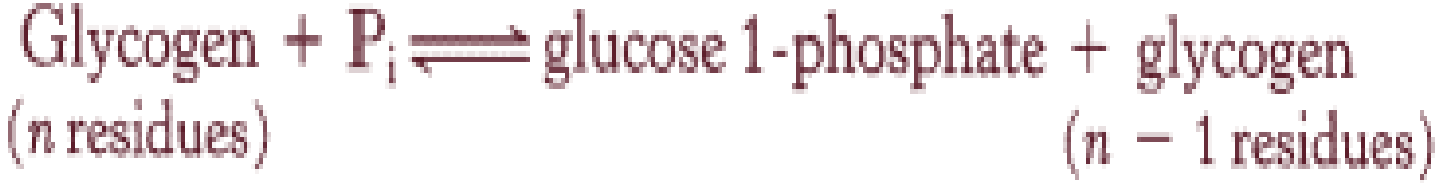
(3) The conversion of glucose 1-phosphate into glucose 6-phosphate for further metabolism.

Enzymes required for glycogen degradation

- The efficient breakdown of glycogen to provide glucose 6-phosphate for further metabolism requires **four enzyme activities**:
- one **to degrade glycogen**,
- two **to remodel glycogen** so that it remains a substrate for degradation, and
- one **to convert the product of glycogen breakdown into a form suitable for further metabolism.**

Enzymes required for glycogen degradation (contd.)

a) Phosphorylase- **Glycogen phosphorylase,** the key enzyme in glycogen breakdown, cleaves its substrate by the addition of orthophosphate (Pi) to yield **glucose 1-phosphate.** The cleavage of a bond by the addition of orthophosphate is referred to as **phosphorolysis.**



Enzymes required for glycogen degradation (contd.)

b) Transferase and Debranching enzyme-

- The **Transferase shifts a block of three glucosyl residues from one outer branch to the other.**

- This transfer exposes a single glucose residue joined by an α -1,6-glycosidic linkage.

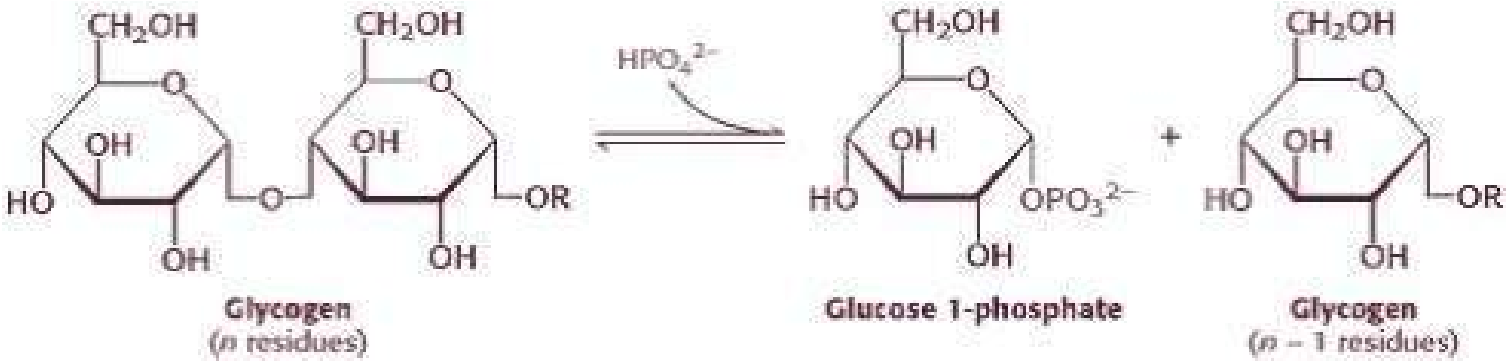
- α-1,6-Glucosidase, also known as the debranching enzyme, hydrolyzes the α -1, 6- glycosidic bond**, resulting in the release of a free glucose molecule.**

Enzymes required for glycogen degradation (contd.)

c) Phosphoglucomutase- **Glucose 1-phosphate** formed in the phosphoroylytic cleavage of glycogen must be **converted into glucose 6-phosphate** to enter the metabolic mainstream. This shift of a phosphoryl group is catalyzed by **Phosphoglucomutase.**

Steps of degradation

1) Release of Glucose-1-P



- Phosphorylase catalyzes the sequential removal of glucosyl residues from the nonreducing ends of the glycogen molecule (the ends with a free 4-OH group.
- Orthophosphate splits the glycosidic linkage between C-1 of the terminal residue and C-4 of the adjacent one.

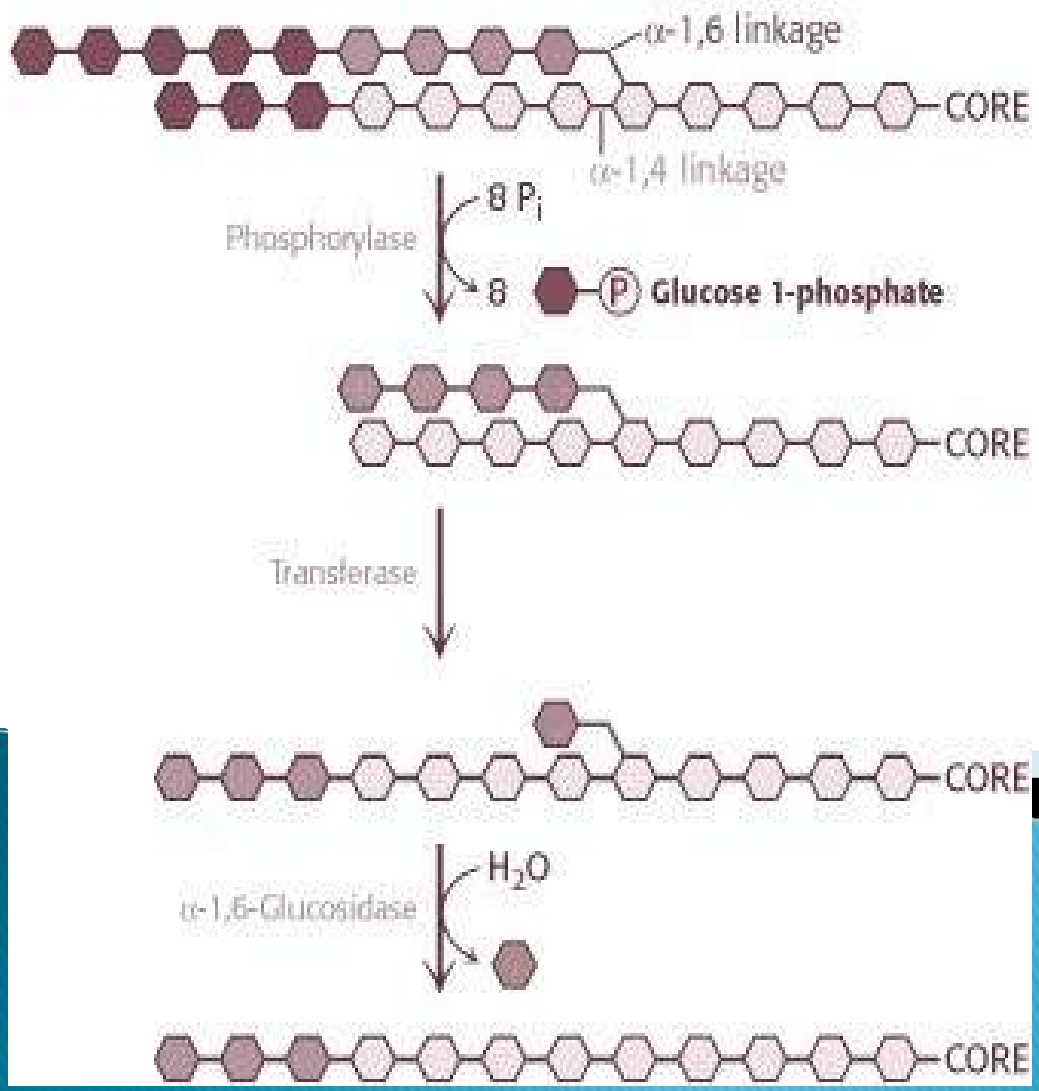
Advantages of phosphoroylytic cleavage

- The phosphoroylytic cleavage of glycogen is energetically advantageous because **the released sugar is already phosphorylated.**
- In contrast, **a hydrolytic cleavage would yield glucose, which would then have to be phosphorylated at the expense of the hydrolysis of a molecule of ATP** to enter the glycolytic pathway.
- An additional advantage of phosphoroylytic cleavage for muscle cells is that **glucose 1-phosphate, negatively charged under physiological conditions, cannot diffuse out of the cell.**

2) Remodeling of the glycogen substrate to permit further degradation

- The **α-1,6-glycosidic bonds** at the branch points are **not susceptible to cleavage by phosphorylase.**
- Glycogen phosphorylase stops cleaving α -1,4 linkages when it reaches a terminal residue four residues away from a branch point.
- Because about 1 in 10 residues is branched, **glycogen degradation by the phosphorylase alone would come to a halt after the release of six glucose molecules per branch.**

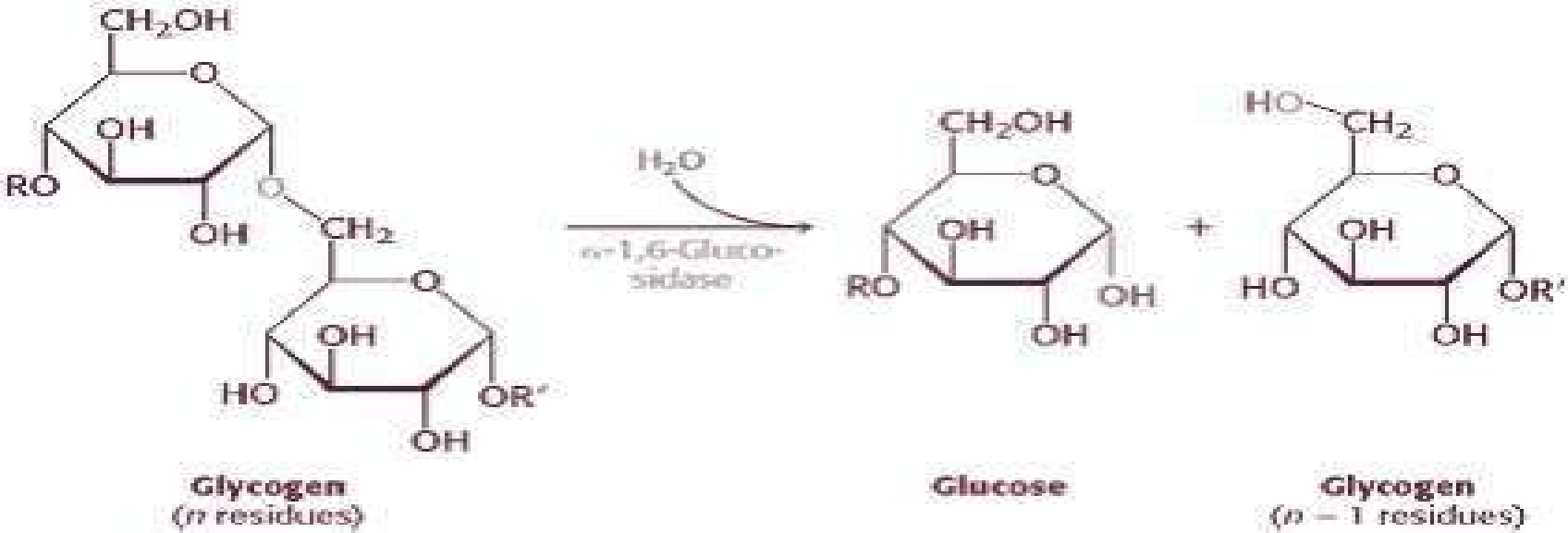
2) Remodeling of the glycogen substrate to permit further degradation(Contd.)



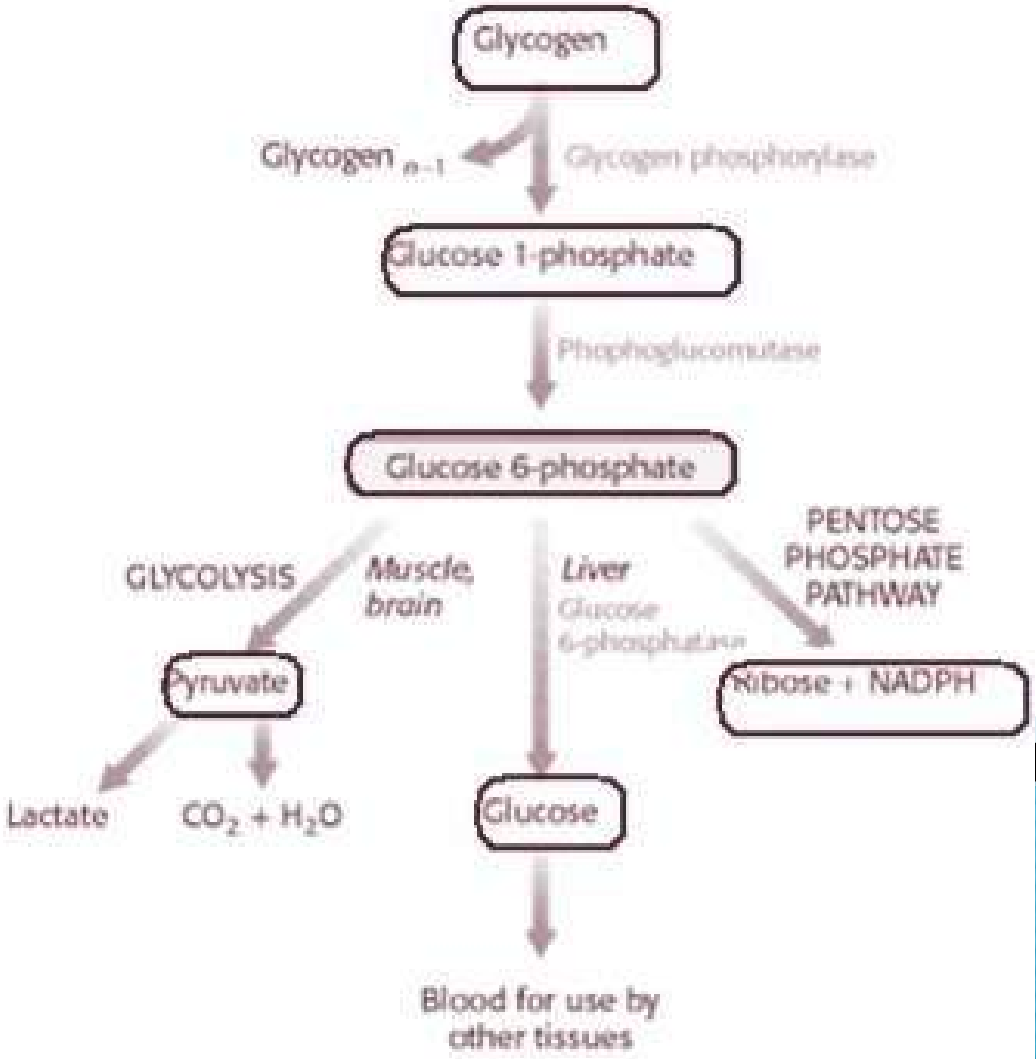
- Transferase shifts a block of three glycosyl residues from one outer branch to the other.
- This transfer exposes a single glucose residue joined by an α-1,6-glycosidic linkage.
- Debranching enzyme, hydrolyzes the α -1, 6- glycosidic bond, resulting in the release of a free glucose molecule.

2) Remodeling of the glycogen substrate to permit further degradation (contd.)

- Transferase and α-1,6-glucosidase, remodel the glycogen for continued degradation by the phosphorylase.
- The free glucose molecule released by the action of debranching enzyme is phosphorylated by the glycolytic enzyme hexokinase.
- Thus, the Transferase and α-1,6-glucosidase convert the branched structure into a linear one, which paves the way for further cleavage by phosphorylase.



3) Conversion of glucose 1-phosphate to glucose 6-phosphate



- Phosphoglucomutase converts glucose 1- phosphate into glucose 6- phosphate in a reversible reaction.
- Glucose 6-phosphate derived from glycogen can (1) be used as a fuel for muscle; (2) be converted into free glucose in the liver and subsequently released into the blood;

(3) be processed by the pentose phosphate pathway.

Conversion of Glucose-6-P to free Glucose

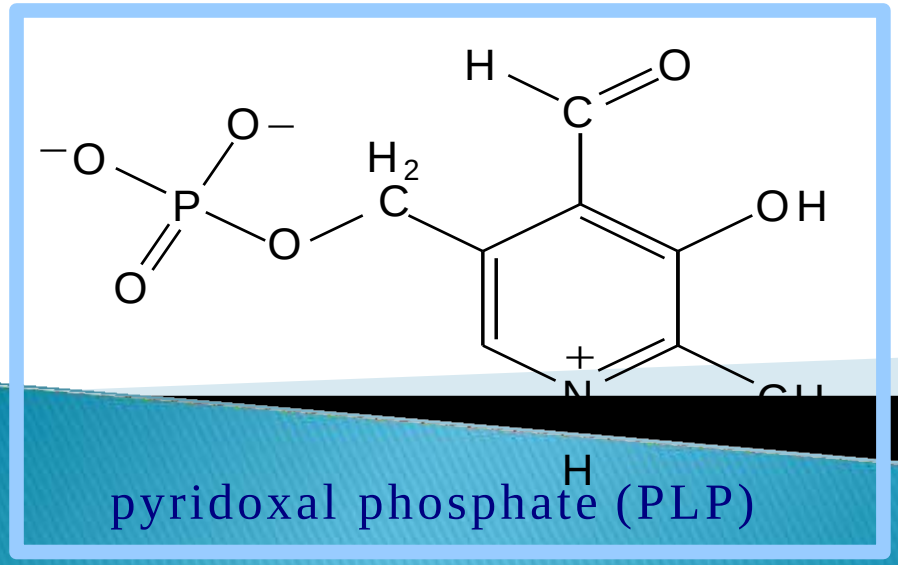
The liver contains a hydrolytic enzyme, **glucose 6-phosphatase,** which cleaves the phosphoryl group to form free glucose and orthophosphate.



- **Glucose 6-phosphatase is absent from most other tissues**
- Consequently, glucose 6-phosphate is retained for the generation of ATP.
- The liver releases glucose into the blood during muscular activity and between meals to be taken up primarily by the brain and skeletal muscle.

Role of Pyridoxal –P in Glycogen degradation

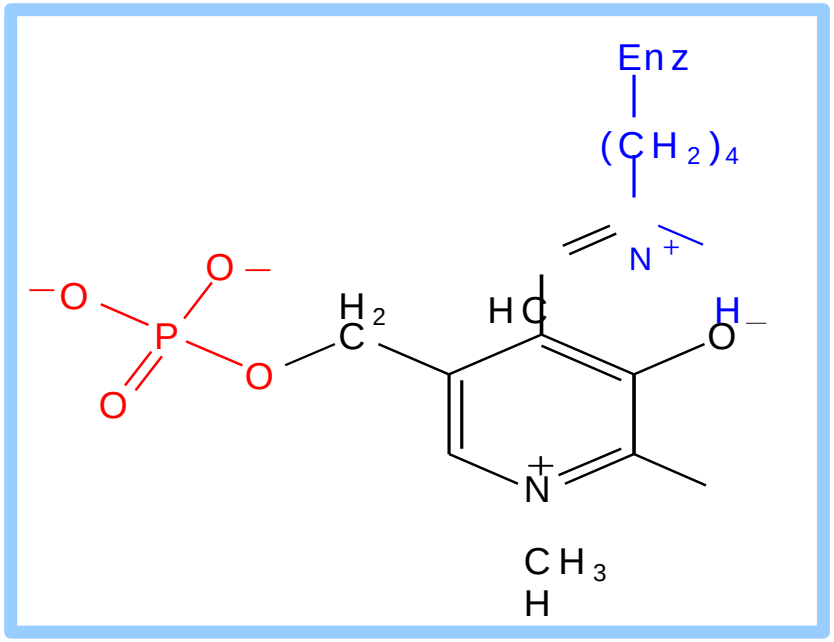
Pyridoxal phosphate (PLP), a derivative of vitamin B_6, serves as prosthetic group for Glycogen Phosphorylase.



pyridoxal phosphate (PLP)

Role of Pyridoxal –P in Glycogen degradation(contd.)

- Pyridoxal phosphate (PLP) is held at the active site by a **Schiff base** linkage, formed by reaction of the aldehyde of PLP with the ε- amino group of a **lysine** residue.
- In contrast to its role in other enzymes, the **phosphate of PLP** is involved in acid/base catalysis by Phosphorylase.



Enzyme (Lys)-PLP Schiff base

Energetics of Glycogenesis and Glycogenolysis

- one ATP is hydrolyzed incorporating glucose 6-phosphate into glycogen.
- The energy yield from the breakdown of glycogen is highly efficient.
- About 90% of the residues are phosphorolytically cleaved to glucose 1-phosphate, which is converted at no cost into glucose 6-phosphate.
- The other 10% are branch residues, which are hydrolytically cleaved.
- One molecule of ATP is then used to phosphorylate each of these glucose molecules to glucose 6-phosphate.

Regulation of Glycogenolysis

- Glycogen Synthase and Glycogen Phosphorylase are **reciprocally regulated**, by allosteric effectors and by phosphorylation.
- The control of phosphorylase differs between liver & muscle
- **In the liver the role of glycogen is to provide free glucose** for export to maintain the blood concentration of glucose;
- **In muscle the role of glycogen is to provide a source of glucose 6-phosphate for glycolysis in** response to the need for ATP for muscle contraction.

Regulation of Glycogenolysis by Allosteric modification

- **Glycogen Phosphorylase** in **muscle** is subject to allosteric regulation by **AMP, ATP, and glucose-6-phosphate.**
- A separate isozyme of Phosphorylase expressed in liver is less sensitive to these allosteric controls.
- **AMP (**present significantly when ATP is depleted) **activates** Phosphorylase, promoting the relaxed conformation.
- **ATP** & **glucose-6-phosphate**, which both have binding sites that overlap that of AMP, **inhibit** Phosphorylase

Thus glycogen breakdown is inhibited when ATP and glucose-6-phosphate are plentiful.

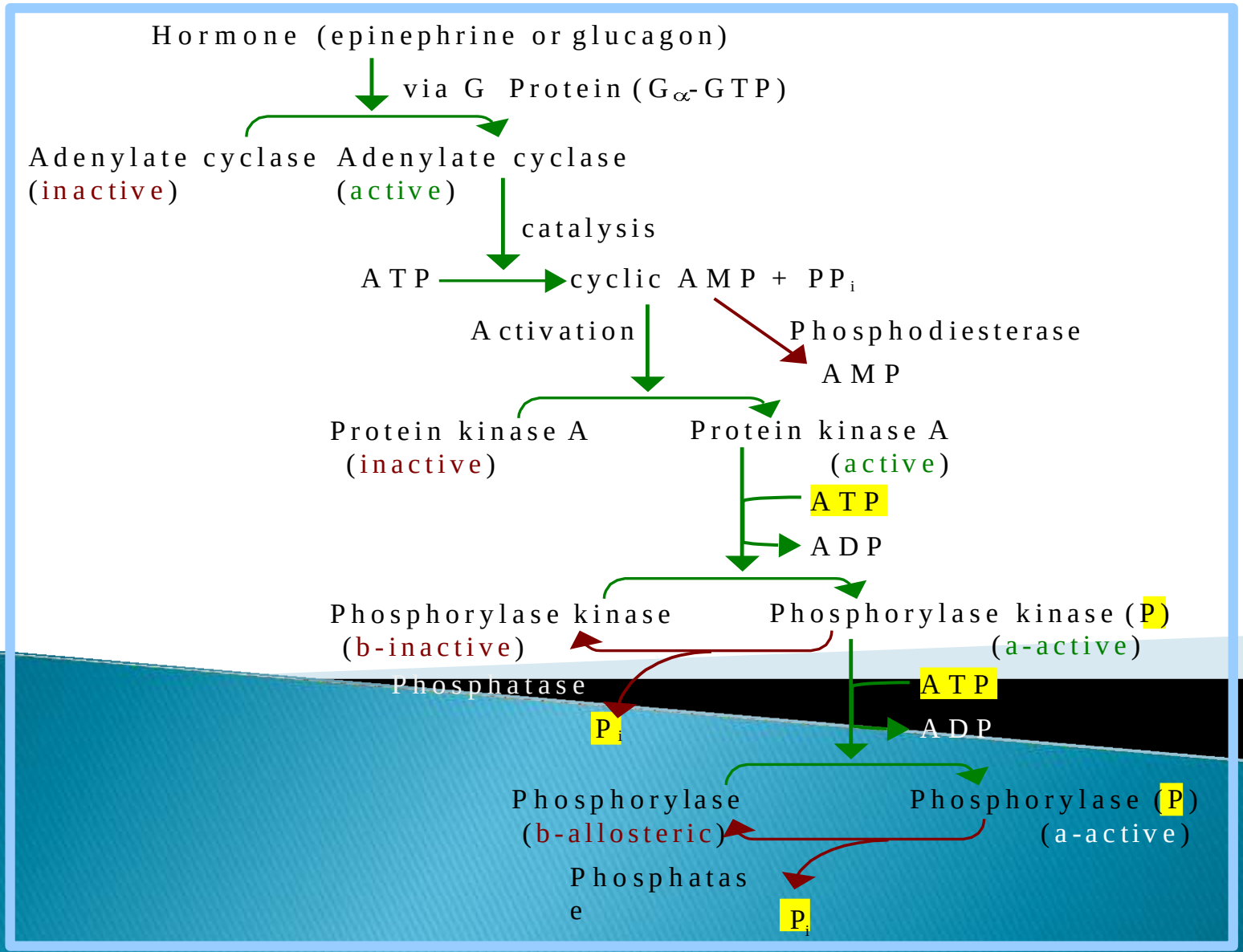
Regulation of Glycogenolysis by Covalent Modification (phosphorylation):

- The cAMP cascade results in **phosphorylation** of a serine hydroxyl of Glycogen Phosphorylase, which promotes transition to the **active** state.
- The phosphorylated enzyme is **less sensitive to allosteric inhibitors**.
- Thus, even if cellular ATP & glucose-6-phosphate are high, Phosphorylase will be active.
- The glucose-1-phosphate produced from glycogen in liver may be converted to free **glucose** for release to the blood.
- With this hormone-activated regulation, the needs of the organism take precedence over needs of the cell.

Regulation of Glycogenolysis by Covalent Modification (phosphorylation):

- The enzyme phosphorylase is activated by phosphorylation catalyzed by **phosphorylase kinase (to yield phosphorylase a)** and
- Inactivated by dephosphorylation catalyzed by **phosphoprotein phosphatase (to yield phosphorylase b)**, in response to hormonal and other signals.

Regulation of Glycogenolysis by Covalent Modification (phosphorylation):



Role of cAMP In Glycogen degradation

- Increasing the concentration of cAMP **activates cAMP-dependent protein kinase,** which catalyzes the phosphorylation by ATP of inactive **phosphorylase kinase b** to active **phosphorylase kinase a,** which in turn, phosphorylates **phosphorylase b to phosphorylase a.**

- In the liver, cAMP is formed in response to glucagon, which is secreted in response to falling blood glucose; muscle is insensitive to glucagon.

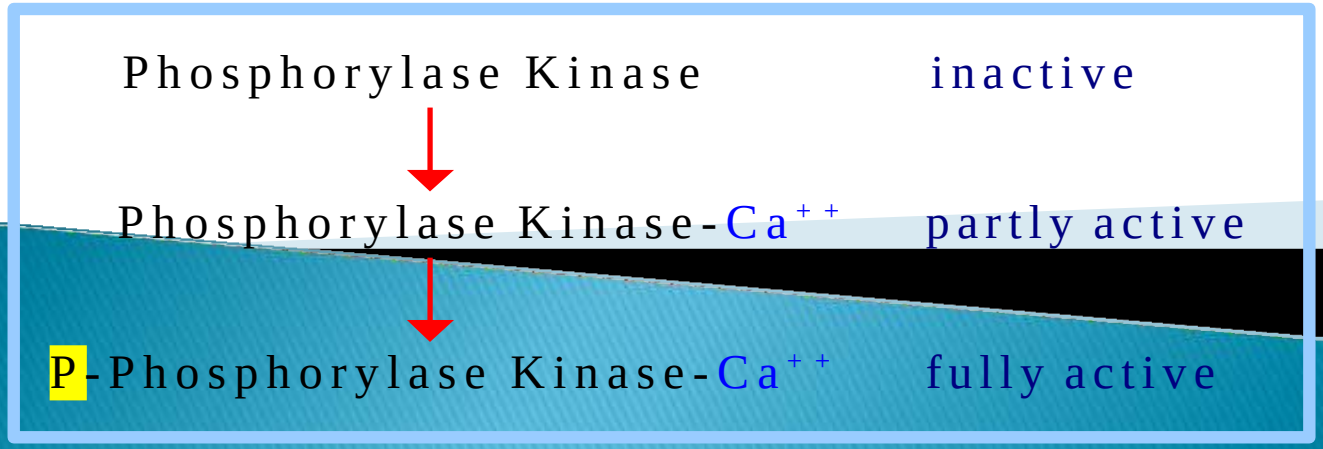
- **In muscle, the signal for increased cAMP formation is the action of norepinephrine,** which is secreted in response to fear or fright, when there is a need for increased glycogenolysis to permit rapid muscle activity.

Role of Ca^{++} in glycogen degradation

- **Ca^{++}** also regulates glycogen breakdown in **muscle**.
- During activation of contraction in skeletal muscle, **Ca^{++}** is released from the sarcoplasmic reticulum to promote actin/myosin interactions.
- The released **Ca^{++}** also **activates Phosphorylase Kinase**, which in muscle includes **calmodulin** as its **δ subunit**.
- Phosphorylase Kinase is partly activated by binding of **Ca^{++}** to this subunit.

Role of Ca^{++} in glycogen degradation

- Muscle phosphorylase kinase, which activates glycogen phosphorylase, is a tetramer of four different subunits-**α, β ,Y and δ .**
- The α and β subunits contain serine residues that are phosphorylated by cAMP-dependent protein kinase. The δ subunit is identical to the Ca^{2+}-binding protein **calmodulin.**
- The binding of Ca^{2+} activates the catalytic site of the subunit even while the enzyme is in the dephosphorylated b state; the phosphorylated a form is only fully activated in the presence of Ca^{2+}.



Role of Insulin in Glycogen degradation

- Both phosphorylase a and phosphorylase kinase a are dephosphorylated and inactivated by **protein phosphatase-1.**
- Protein phosphatase-1 is inhibited by a protein, **inhibitor-1,** which is active only after it has been phosphorylated by cAMP-dependent protein kinase.
- Thus, cAMP controls both the activation and inactivation of phosphorylase.
- **Insulin reinforces this effect by inhibiting the activation of phosphorylase b.**
- It does this indirectly by increasing uptake of glucose, **leading to increased formation of glucose 6-phosphate, which is an inhibitor of phosphorylase kinase.**

Reciprocal regulation of Glycogen synthesis and degradation

- **Glycogen Synthase & Phosphorylase activity are reciprocally regulated**
- At the same time as phosphorylase is activated by a rise in concentration of cAMP (via phosphorylase kinase), glycogen synthase is converted to the inactive form;
- both effects are mediated via **cAMP-dependent protein kinase .**
- Thus, inhibition of glycogenolysis enhances net glycogenesis, and inhibition of glycogenesis enhances net glycogenolysis

Glycogen Metabolism(Summary)

- Glycogen represents the **principal storage** form of carbohydrate in the body, mainly in the **liver and muscle.**
- Glycogen is synthesized from glucose by the pathway of glycogenesis.
- It is broken down by a separate pathway, glycogenolysis. Glycogenolysis leads to glucose formation in liver and lactate formation in muscle owing to the respective presence or absence of glucose 6-phosphatase.
- **Cyclic AMP integrates the regulation of glycogenolysis and glycogenesis** by promoting the simultaneous activation of phosphorylase and inhibition of glycogen synthase.
- **Insulin acts reciprocally by inhibiting glycogenolysis and stimulating glycogenesis.**
- Inherited deficiencies in specific enzymes of glycogen metabolism in both liver and muscle are the causes of glycogen storage diseases.

REFERENCES

- Harper
- Lippincott